

Mohammad Yousuf Qureshi

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EDUCATION

Georgia Institute of Technology – [M.S. Analytics](#)

May 2026

Computational Data Analytics Track, CGPA: 3.83 / 4.0

Machine Learning, Time Series Analysis, Data & Visual Analytics, Regression, Bayesian Statistics, NLP, Database Systems, Data Analytics in Business

Texas A&M University at Qatar – [B.S. Mechanical Engineering](#)

May 2020

Magna Cum Laude, CGPA 3.74 / 4.0 | Tau Beta Pi | Pi Tau Sigma | Dean's Honor Roll

ANALYTICS PROJECTS

Social Media Signals for Flight Booking Surges – Lufthansa Group

NDA Industry Project | github.com/YousufQ9/reddit-flight-demand-signals

- Led social media analysis pipeline on ~45M rows of proprietary booking data; processed unstructured data from 5 travel subreddits using VADER sentiment, spaCy NER, and RapidFuzz fuzzy matching to map 221 destinations to ISO codes
- Discovered ~60% of raw Reddit–booking correlation was spurious seasonality; applied STL deseasonalization to isolate genuine 7–28-day leading signals; Granger causality confirmed in 7 of 41 markets; final relevance score achieved AUC = 0.842 on high-signal markets and 9× lift in booking spike prediction

Clinical Chest X-Ray Report Generation – LSTM vs. Transformer

github.com/YousufQ9/dl-cxr-report-gen

- Implemented hierarchical LSTM baseline (Jing et al. 2018) with DenseNet-121 encoder achieving BLEU-1 of 0.286 (+33% over default); built full evaluation module covering linguistic (BLEU, METEOR, CIDEr, ROUGE-L) and clinical metrics (CheXbert F1, hallucination rate)
- Clinically guided Transformer with weighted cross-entropy loss improved CheXbert macro-F1 by 23% and CIDEr by 43% over vanilla Transformer, confirming a measurable fluency–clinical accuracy trade-off

Inflationary Impact of U.S. Tariffs – Interactive Dashboard

github.com/YousufQ9/tariff_impact_dashboard

- Built ARIMAX models (with tariff rates as exogenous variable) for electronics, food & apparel CPI forecasting using 10 years of BLS/FRED/USITC data; deployed results to an interactive Tableau CPI Simulator with 130+ public views, rated 4.56/5 by users

Dog Sanctuary Management System – Database Application

github.com/YousufQ9/database_stevies_dog_sanctuary

- Designed and built a full-stack web application (React frontend, Django backend, MySQL database) for a fictional dog rescue organization, covering the complete lifecycle from requirements analysis to ER modeling, relational schema design, and implementation
- Implemented role-based access control (Volunteer vs. Executive Director permissions) with session-based validation at the application and query level, and modeled complex business rules directly into the schema, including tiered adoption fee calculations and duplicate-expense prevention constraints

EXPERIENCE

Estimation Engineer – Fire Protection Systems · JBK Controls, Qatar

Sep 2020 – May 2024

Sprinkler Systems, Deluge Systems, Pre-Action Systems, Foam Systems, Clean Agent (FM200 & NOVEC) Systems)

- Forecasted material and labor costs for fire suppression projects using Excel-based cost models, referencing historical project data and supplier pricing to inform estimates.
- Extracted and structured quantity take-off data from AutoCAD drawings into detailed Excel cost templates
- Managed supplier negotiations and site assessments across sprinkler, foam, and clean-agent (FM200/NOVEC) systems
- Analyzed and compared multi-vendor supplier quotations to identify cost-optimal procurement options during negotiations, balancing pricing against technical compliance requirements.

TECHNICAL SKILLS

Languages & Tools: Python, R, SQL, Tableau, Power BI, Git, Jupyter, Docker, Excel, Data Visualization, Streamlit

Cloud & Big Data: PySpark, Apache Spark (DataFrame API), Databricks, AWS (S3, Athena), Google Cloud Platform

ML & Analytics: Scikit-learn, Statsmodels, ARIMAX, XGBoost, Random Forest, Clustering, Regression, Bayesian inference, Matplotlib, Seaborn

Data Engineering: Pandas, NumPy, MySQL, feature engineering, data cleaning & imputation, ETL pipelines, Data modelling, Schema design

NLP: VADER, spaCy, RapidFuzz, NER, text preprocessing

CERTIFICATIONS

Microsoft Azure Essentials – Professional Certificate, Microsoft & LinkedIn Learning

Career Essentials in GitHub – Professional Certificate, GitHub & LinkedIn Learning

Career Essentials in Generative AI – Microsoft & LinkedIn Learning

Data Analytics Job Simulation – Deloitte (via Forage)